Manual

Cockpit for System Administrators

SIS-CSA 3.0/3.1

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Contents

1 Cockpit for System Administrators

Overview

Purpose

The Cockpit for System Administrators allows for an automated notification based on operation statuses of the HYDRA server.

Implementation notes

You use the Cockpit for System Administrators if you wish to receive system-supported warning messages about specific operating statuses of the HYDRA server in order to be able to react in good time.

Integration

The Cockpit for System Administrators provides escalations which can be used within Escalation Management.

Features

System Integration Service (SIS) for the automatic monitoring of:

- Fill levels of the HYDRA database
- Fill levels of the MES server drives

with adjustable limit values. Alert messages for a definable group of employees by e-mail when limit values are exceeded.

2 Escalations for System Administrators

Event	Description	Identifiers	Description	Please note
TNR.OFFLINE	Terminal is offline	TNR.TNR	Terminal number	The event is triggered if the duration since the last status posting of the terminal has exceeded the target cycle of status postings.
		TNR.BEZL	Terminal description	
		TNR.BEZK	Terminal location	
		TNR.ZYKL:I	Time passed since the last status posting in seconds	
		TNR.ZYKL:S	Target posting cycle in seconds	
DB.INCREMENT_TOO_LARGE	Fill level of table spaces too high	DB.INC:GR	Increase in percent	The event is triggered if the fill level of individual files/table spaces has increased by <DB.INC:GR> within the past few days.
		DB.INC:TG	Period considered	
FILESYS.FILL_LEVEL_EXCEEDED	Fill level of drive exceeded	FILESYS.FILESYS	File system	
		FILESYS.SUM	File system size	
		FILESYS.SUM:EINH	Unit of file size	
		FILESYS.FREE	Free memory	
		FILESYS.FREE:EINH	Unit of free memory	
		FILESYS.FREE:PROZ	Free memory in percent	
		FILESYS.USED	Used memory	
		FILESYS.USED:EINH	Unit of used memory	
		FILESYS.USED:PROZ	Used memory in percent	
		FILESYS.USEDGR:PROZ	Limit in percent	
DB.FILL_LEVEL_EXCEEDED	Fill level of database file groups/table spaces exceeded	DB.DBSPACE	File group / table space	The event is triggered if the fill level of one or more table spaces has exceeded the value <DB.USEDGR:PROZ>.
		DB.DBSNUM	Number (Informix)	
		DB.NCHUNKS	Number of chunks (Informix)	
		DB.SUM	Size	
		DB.SUM:EINH	Unit of size	
		DB.FREE	Free memory	
		DB.FREE:EINH	Unit of free memory	
		DB.FREE:PROZ	Free memory in percent	
		DB.USED	Used memory	
		DB.USED:EINH	Unit of used memory	
		DB.USED:PROZ	Used memory in percent	
		DB.USEDGR:PROZ	Limit in percent	

3 Cockpit for System Administrators

3.1 Requirements

The system provides the option to output metrics to the monitoring and alert system Prometheus. You can visualize these metrics using the analytics system Grafana.

In combination with other data suppliers providing for example metrics of the operating system and the database management system, this solution can be used to monitor whether the system operates well; in case of problems, you can be informed.

To output metrics you need the **SIS-CSA** license and **Service Pack 15**.

"Prometheus" and "Grafana" are products of third party manufacturers. Both products are neither sold nor supported by MPDV.

For Prometheus MPDV only provides the metrics for query.

For Grafana MPDV provides exemplary dashboards.

3.2 Activation of metrics output

By default, the output of metrics is deactivated. Make the configuration below to enable the metrics output:

<*InstallDir*>\WSP<*SystemNo*>\config\application.properties
`management.security.enabled=true`

3.3 Prometheus installation

Find an overview of Prometheus here:
https://prometheus.io/docs/introduction/overview/

Install Prometheus. Find instructions on the installation, operation and configuration of the product here:
https://prometheus.io/docs/prometheus/latest/getting_started

Configuration of the Exporter

Extend the Prometheus configuration as follows to have Prometheus query the metrics:

<*Prometheus folder*>\prometheus.yml

```
scrape_configs:
  - job_name: '<name of configuration>'
    scheme: 'https'
    metrics_path: '/prometheus'
    static_configs:
    - targets: ['<host name of server>:<port>']
```

The port depends on the system queried:

- System 1: **8080**
- System 2: **8081**
- System 3: **8082**
- etc.

Installation of WMI Exporter / Node Exporter

The following Prometheus exporters can be installed to output metrics of the operating system:

- Windows: https://github.com/martinlindhe/wmi_exporter
- LINUX: https://github.com/prometheus/node_exporter

Find notes on the configuration of the adapter here:
https://prometheus.io/docs/introduction/first_steps/

Use at least the following start parameters of the WMI Exporter for the basic functions:

```
--log.format logger:eventlog?name=wmi_exporter
--telemetry.addr :9182
```

If the Microsoft SQL Server is used as database and if this server is on the same host as the system, you can additionally use the following start parameters:

```
--collectors.enabled cpu,cs,logical_disk,net,os,system,mssql,textfile
--collectors.mssql.classes-enabled=accessmethods,bufman,databases,genstats,locks,memmgr,sqlstats
```

MPDV Index dashboard, tab *SQL Server:* Only if the above parameters are set, the panel functions of the dashboard provided by MPDV are active.

Checking the result

After start of Prometheus, enter the following address in a browser:
http://***<name of Prometheus host>***:9090

The relevant web page must then be displayed. In the provided selection of metrics, the metrics of the following naming scheme must be available shortly after: `wsp_services_*` (e.g. `wsp_services_duration_total_seconds_sum`).

3.4 Grafana installation

Find an overview of Grafana here:
https://grafana.com/

Install Grafana. Find instructions on the installation, operation and configuration of the product here:
https://grafana.com/docs/installation/

Configuration of Prometheus as data source

After start of Grafana, enter the following address in a browser:
http://***<name of Grafana host>***:3000
The relevant web page must then be displayed.

Open the page to configure the data sources. Add Prometheus as data source.
All steps required to visualize system metrics using Grafana are then fulfilled.

Loading example dashboards

MPDV provides some example dashboards. You can download them from the page below and integrate them in your Grafana installation. Use the search function of this page and search for "MPDV".
https://grafana.com/grafana/dashboards

To use the MPDV dashboards, install further Grafana panel plug-ins, if required. The requirements of an MPDV panel are listed in the *Dependencies.*
Find the installation instruction for the plug-ins here:
https://grafana.com/docs/plugins/installation/

4 Prometheus System Metrics

4.1 Naming

The general rules normally apply for the naming of metrics and labels. The following websites provide a very good overview:

- https://prometheus.io/docs/practices/naming/
- https://prometheus.io/docs/concepts/data_model/#metric-names-and-labels

In addition to the general rules, the following naming conventions apply:

- Metrics, which are centrally available for 100 % of all available services, start with **`wsp.services.`**….
  Metrics with this name are exclusively provided by the basic system functions.
- Metrics provided for a subset of services start with **`wsp.services.subset.`**….
  Only use this naming for metrics if their evaluation is useful for all services.
- Metrics only provided for specific services start with **`wsp.service.`*`<service name in lower case>`*`.`**….
  The relevant service documentation describes the relevant labels.

4.2 Metric data types

The following data types are generally available for metrics:

- Summary – is used to total values
  (Example: number of processed data records, duration of processing)
  For each summary metric, the following parameters are provided:
  - ***`<name of metric>`*`_sum`**
  - ***`<name of metric>`*`_count`**
  - ***`<name of metric>`*`_max`**
- Counter – Metric, which is incremented with each call.
  (Example: number of calls, number or errors)
- Timer – is used to total times (in seconds)
  (Example: duration of a service)
  For each timer metric, the following parameters are provided:
  - ***`<Name of the metric>`*`_seconds`**
- Gauge – current value, not totaled
  (Example: current memory usage, current table space usage, alive status)

General note: Only the currently available content of a metric is transmitted at the time Prometheus collects the data.
Some of the data provided, e.g. with gauge, is not "combined" and transferred one by one. Instead, only the last transferred value (which had overwritten the predecessor values) is transferred.

In Prometheus, the relevant Prometheus configuration distinguishes the different systems. The native labels **`job`** and **`instance`** are provided to distinguish the data source.

4.3 Standard metrics...

...provided by basic functions

The basic functions provide the following "basic metrics" for services:

Metric name with labels	Type	Value
wsp.services.calls.total **{service_domain=”***<service domain>***”,** **service_function=”list|insert|delete|** **update|modify|copy|** **lock|unlock|…”,** **device_id=”***<client device ID>***”,** **user_id=”***<user ID>***”,** **distribution_id=”***<distribution ID>***”,** **processing_result=”processed|failed”** **}**	Count	Counts service calls
wsp.services.errors.total **{service_domain=”***<service domain>***”,** **service_function=”list|insert|delete|** **update|modify|copy|** **lock|unlock|…”,** **device_id=”***<client device ID>***”,** **user_id=”***<user ID>***”,** **distribution_id=”***<distribution ID>***”,** **processing_result=”processed|failed”** **}**	Count	Counts how often a service is canceled with error
wsp.services.duration.total_seconds **{service_domain=”<service domain>”,** **service_function=”list|insert|delete|** **update|modify|copy|**	Timer	Processing time of service in seconds

lock|unlock|…", device_id="<client device ID>", user_id="<user ID>", distribution_id="<distribution ID>", processing_result="processed|failed" }		
wsp.services.returned_records.total_count/_max/_sum {service_domain="<service domain>", service_function="list|insert|delete| update|copy| lock|unlock|…", device_id="<client device ID>", user_id="<user ID>", distribution_id="<distribution ID>", resultset_id="<result set ID>" processing_result="processed| empty|failed|…", }	Summary	Number of data records in the result sets of the highest level that the service returns to the calling process.

… provided by services

The following metrics with a global naming for all services can be provided by specific services:

Metric name with labels	Type	Value
wsp.services.subset.records.total_count/_max/_sum {service_domain="<service domain>", service_function="list|insert|delete| update|copy| lock|unlock|…", device_id="<client device ID>", user_id="<user ID>", distribution_id="<distribution ID>", processing_result="processed| empty|failed|…", data_type="operation_logon_period| process_value…" record_type="deleted|inserted|	Summary	Number of data records

`updated|listed|`
`remaining|…"`
`}`		

`wsp.services.subset.files.total_count/_max/_sum`
`{service_domain="<service domain>",`
`service_function="list|insert|delete|`
`update|copy|`
`lock|unlock|…",`
`device_id="<client device ID>",`
`user_id="<user ID>",`
`distribution_id="<distribution ID>",`
`processing_result="processed|`
`empty|failed|…",`
`data_type="operations|…"`
`file_type="processed|remaining|…"`
`}`
	`Summary`	Number of files

`wsp.services.subset.time_period.total_count/_max/_sum`
`{service_domain="<service domain>",`
`device_id="<client device ID>",`
`user_id="<user ID>",`
`distribution_id="<distribution ID>",`
`processing_result="processed|`
`empty|failed|…",`
`period_type="processed|remaining|…"`
`}`
	`Summary`	Period of time

For a precise listing of the metrics specifically provided by the services, refer to section "Metrics of specific services".

4.4 Standard labels

service_domain

This label is used to narrow down the analysis, e.g. to a specific service domain. With standard metrics, the service domain is included in a label and not in a metric name. For this reason, you can optionally perform analyses for all services, e.g. to evaluate the number of inserted data records from all services.
This label is always directly created from the service name (character string before last dot).
The metrics, which include the service name already in the metric name, usually do not use this label.

In combination with the label **service_function,** you can filter by a specific service.

service_function

You can use this label to filter services with similar functions. For example, using this label you can make evaluations of all delete services for the runtime. This label is always directly created from the service name (character string after last dot).
The metrics, which include the service name already in the metric name, usually do not use this label.

In combination with the label **service_domain,** you can filter by a specific service.

device_id

Using this label, you can identify for analyses, which client made the request. For example, this can be helpful if several users work on different devices with the same user name.

user_id

Using this label, you can make analyses for the query of specific users. Optionally, you can also compare the queries of different users.

distribution_id

You use this label to distinguish several parallel services having the same task. You usually use these parallel services to distribute the load.

The content of this label is usually directly derived from the service parameter **`distribution.id`**.

resultset_id

If a service returns several different result sets to the client, you can use this label to identify for each result set how many data records it includes.

processing_result

You use this label as additional filter to identify processings that have actually processed data.

Valid values:

- **`processed`**
- **`empty`**
- **`failed`**

Specific services can also use other values.

data_type

If metrics with the same name are output for different data in a service, then you can use this label to identify the data the metric refers to.

Example: Process data statistics are generated for each operation logon period. When the metrics for the number of "relevant" data records are output, you can use the label **data_type** to distinguish the following metrics:

a)
```
wsp.services.subset.records.total_count/_max/_sum
{…,
 data_type="OperationLogonPeriods"
 record_type="processed"
}
```
⇨ Number of processed operation logon periods

b)
```
wsp.services.subset.records.total_count/_max/_sum
{…,
 data_type="OperationLogonPeriods"
 record_type="remaining"
}
```
⇨ Number of the operation logon periods that must still be processed (later).

c)
```
wsp.services.subset.records.total_count/_max/_sum
{…,
 data_type="ProcessValueStatistics"
 record_type="compressed"
}
```
⇨ Number of compressed process parameter statistics

If a service includes only metrics of one kind of data, then this label is not needed.

In the context of other metrics, you can use this label for similar purposes. For example, with file-based processings you can distinguish the different file types.

record_type

You can use this label to evaluate for all services how many data records were created, deleted, changed, for example.

Valid values:

- **`deleted`**
- **`inserted`**
- **`updated`**

- `listed`

Specific services can also use other values.

file_type

You can use this label to distinguish the different types of files that might be relevant in the context of a service call.

Valid values:

- `processed`
- `remaining`

Specific services can also use other values.

4.5 Metrics of specific services

4.5.1 AnalysisOperationLogonPeriods.importFromManufacturing

The following service-specific metrics are provided by the service **AnalysisOperationLogonPeriods.importFromManufacturing**:

Metric name with labels	Type	Value
`wsp.services.subset.records.total_count/_max/_sum {service_domain="AnalysisOperationLogonPeriods", service_function="importFromManufacturing", distribution_id="", processing_result="processed|empty|failed", device_id="<client device ID>", user_id="<user ID>", data_type="OperationLogonPeriods", record_type="processed" }`	`Summary`	Number of processed data records If the service is finished with an error, then the output is `0`. In this case, the label `processing_result` gets the value `failed`. If no data records are imported, the label `processing_result` gets the value `empty`, otherwise `processed`.

4.5.2 ProcessDataSpecifications.batchInsert

The following service-specific metrics are provided by the service **ProcessDataSpecifications.batchInsert**:

Metric name with labels	Type	Value
wsp.services.subset.records.total_count/_max/_sum **{** **service_domain="ProcessDataSpecifications",** **service_function="batchInsert",** **distribution_id="***<content of the (optional) service parameter **distribution.id**>*",** **processing_result="processed|failed",** **device_id="***<client device ID>*",** **user_id="***<user ID>*",** **data_type="ProcessDataSpecifications",** **record_type="inserted"** **}**	`Summary`	Number of processed data records If the service is finished with an error, then the output is `0`. Otherwise, the number of data records added to the staging table is output. In case of an error, the label **`processing_result`** gets the value **`failed`**, otherwise **`processed`**.

4.5.3 ProcessDataSpecifications.distributeStandard

The following service-specific metrics are provided by the service **ProcessDataSpecifications.distributeStandard**:

Metric name with labels	Type	Value

<table>
<tr>
<td>wsp.services.subset.records.total_count/_max/_sum
{service_domain="ProcessDataSpecifications",
service_function="distributeStandard",
distribution_id="<from variable distributionId>",

processing_result="processed|empty|failed",
device_id="<client device ID>",
user_id="<user ID>",
data_type="ProcessDataSpecifications",
record_type="processed"
}</td>
<td>Summary</td>
<td>Number of processed data records

If the service is finished with an error, then the output is 0. In this case, the label processing_result gets the value failed.

If no data records are processed, the label processing_result gets the value empty, otherwise processed.</td>
</tr>
<tr>
<td>wsp.services.subset.records.total_count/_max/_sum
{service_domain="ProcessDataSpecifications",
service_function="distributeStandard",
distribution_id="<from variable distributionId>",

processing_result="processed|empty|failed",
device_id="<client device ID>",
user_id="<user ID>",
data_type="ProcessDataSpecifications",
record_type="remaining"
}</td>
<td>Summary</td>
<td>Number of specification changes waiting to be distributed.

If the service is finished with an error, the label processing_result gets the value failed. If no data records were processed, the label processing_result gets the value empty, otherwise processed.</td>
</tr>
</table>

4.5.4 ProcessDataSpecifications.insert

The following service-specific metrics are provided by the service **ProcessDataSpecifications.insert**:

Metric name with labels	Type	Value

`wsp.services.subset.records.total_count/_max/_sum { service_domain="ProcessDataSpecifications", service_function="insert", distribution_id="<content of the (optional) service parameter distribution.id>", processing_result="processed|failed", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessDataSpecifications", record_type="inserted" }`	`Summary`	Number of processed data records If the service is finished with an error, then the output is `0`, otherwise it is `1`. In this case, the label **`processing_result`** gets the value **`failed`**, otherwise **`processed`**.

4.5.5 ProcessDataSpecifications.aggregatePeriodicStandard

The following service-specific metrics are provided by the service **ProcessDataSpecifications.aggregatePeriodicStandard**:

Metric name with labels	Type	Value
`wsp.services.subset.records.total_count/_max/_sum {service_domain="ProcessDataSpecifications", service_function="aggregatePeriodicStandard ", distribution_id="", processing_result="processed|empty|failed", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessDataSpecifications", record_type="processed" }`	`Summary`	Number of processed data records If the service is finished with an error, then the output is `0`. In this case, the label **`processing_result`** gets the value **`failed`**. If no data records are processed, the label **`processing_result`** gets the value **`empty`**, otherwise **`processed`**.

`wsp.services.subset.records.total_count/_max/_sum {service_domain="ProcessDataSpecifications", service_function="aggregatePeriodicStandard ", distribution_id="", processing_result="processed|empty|failed", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessDataSpecifications", record_type="compressed" }`	Summary	Number of processed data records If the service is finished with an error, then the output is `0`. In this case, the label **`processing_result`** gets the value **`failed`**. If no data records are processed, the label **`processing_result`** gets the value **`empty`**, otherwise **`processed`**.
`wsp.services.subset.time_period.total.seconds_count/_max/_sum {service_domain="ProcessDataSpecifications", service_function="aggregatePeriodicStandard ", distribution_id="", processing_result="processed|empty|failed", device_id="<client device ID>", user_id="<user ID>", period_type="processed" }`	Summary	Period of time that has been aggregated (in seconds) If the service is finished with an error, then the output is `0`. In this case, the label **`processing_result`** gets the value **`failed`**. If no data records are processed, the label **`processing_result`** gets the value **`empty`**, otherwise **`processed`**.
`wsp.services.subset.time_period.total.seconds_count/_max/_sum {service_domain="ProcessDataSpecifications", service_function="aggregatePeriodicStandard ", distribution_id="",`	Summary	Period of time until offset that must still be aggregated (in seconds) If the service is finished with an error, then the output is `0`. In this case, the label **`processing_result`** gets the

processing_result="processed|empty|failed", **device_id="***<client device ID>***",** **user_id="***<user ID>***",** **period_type="remaining"** **}**		value **failed**. If no data records are processed, the label **processing_result** gets the value **empty**, otherwise **processed**.

4.5.6 ProcessDataStatistics.aggregateByOperation

The following service-specific metrics are provided by the service **ProcessDataStatistics.aggregateByOperation**:

Metric name with labels	Type	Value
wsp.services.subset.records.total_count/_max/_sum **{service_domain="ProcessDataStatistics",** **service_function="aggregateByOperation",** **distribution_id="***<from variable distributionId>***",** **processing_result="processed|empty|failed",** **device_id="***<client device ID>***",** **user_id="***<user ID>***",** **data_type="OperationLogonPeriods",** **record_type="processed"** **}**	Summary	Number of processed operation logon periods If the service is finished with an error, then the output is 0. In this case, the label **processing_result** gets the value **failed**. If no data records are processed, the label **processing_result** gets the value **empty**, otherwise **processed**.

wsp.services.subset.records.total_count/_max/_sum **{service_domain="ProcessDataStatistics",** **service_function="aggregateByOperation",** **distribution_id="***<from variable distributionId>***",** **processing_result="processed|empty|failed",** **device_id="***<client device ID>***",** **user_id="***<user ID>***",** **data_type="OperationLogonPeriods",** **record_type="remaining"** **}**	Summary	Number of operation logon periods that must still be processed (until specified offset). If the service is finished with an error, the label **processing_result** gets the value **failed**. If no data records are processed, the label **processing_result** gets the value **empty**, otherwise **processed**.
wsp.services.subset.records.total_count/_max/_sum **{service_domain="ProcessDataStatistics",** **service_function="aggregateByOperation",** **distribution_id="***<from variable distributionId>***",** **processing_result="processed|empty|failed",** **device_id="***<client device ID>***",** **user_id="***<user ID>***",** **data_type="ProcessValueStatistics"** **record_type="compressed"** **}**	Summary	Number of processed data records If the service is finished with an error, then the output is 0. In this case, the label **processing_result** gets the value **failed**. If no data records are processed, the label **processing_result** gets the value **empty**, otherwise **processed**.

4.5.7 ProcessValues.aggregateCurveProgression

The following service-specific metrics are provided by the service **ProcessValues.aggregateCurveProgression**:

Metric name with labels	Type	Value

wsp.service.processvalues.aggregatecurve progression.compression.records.total_count/_max/_sum **{distribution_id=**"*<content of the (optional) service parameter* ***distribution.id***>", **processing_result=**"**processed|empty|failed**", **device_id=**"*<client device ID>*", **user_id=**"*<user ID>*", **compression=**"*<target table of compression>*", **record_type=**"**processed**" **}**	Summary	Number of processed raw data records If the service is finished with an error, then the output is 0. In this case, the label **processing_result** gets the value **failed**. This parameter is written for each table that must be aggregated (with changing label **compression**). It is globally written for workplace and process parameters of the pool of orders. If no data records are processed, the label **processing_result** gets the value **empty**, otherwise **processed**.
wsp.service.processvalues.aggregatecurve progression.compression.records.total_count/_max/_sum **{distribution_id=**"*<content of the (optional) service parameter* ***distribution.id***>", **processing_result=**"**processed|empty|failed**", **device_id=**"**<client device ID>**", **user_id=**"**<user ID>**", **compression=**"**<target table of compression>**", **record_type=**"**compressed**" **}**	Summary	Number of aggregated data records. If the service is finished with an error, then the output is 0. In this case, the label **processing_result** gets the value **failed**. This parameter is written for each table that must be aggregated (with changing label **compression**). It is globally written for workplace and process parameters of the pool of orders.

		If no data records are processed, the label **`processing_result`** gets the value **`empty`**, otherwise **`processed`**.

4.5.8 ProcessValues.batchInsert

The following service-specific metrics are provided by the service **ProcessValues.batchInsert**:

Metric name with labels	Type	Value
`wsp.services.subset.records.total``_count/_max/_sum` **`{`** **`service_domain="ProcessValues",`** **`service_function="batchInsert",`** **`distribution_id="`***`<content of the (optional) service parameter`* ***`distribution.id`***`>",` **`processing_result="processed|failed",`** **`device_id="`***`<client device ID>`*`",` **`user_id="`***`<user ID>`*`",` **`data_type="ProcessValues",`** **`record_type="inserted"`** **`}`**	`Summary`	Number of processed data records If the service is finished with an error, then the output is `0`. In case of an error, the label **`processing_result`** gets the value **`failed`**, otherwise **`processed`**.

4.5.9 ProcessValues.distributeStandard

The following service-specific metrics are provided by the service **ProcessValues.distributeStandard**:

Metric name with labels	Type	Value

wsp.services.subset.records.total_count/_max/_sum **{service_domain="ProcessValues",** **service_function="distributeStandard",** **distribution_id="***<from variable distributionId>***",** **processing_result="processed|empty|failed",** **device_id="***<client device ID>***",** **user_id="***<user ID>***",** **data_type="ProcessValues",** **record_type="processed"** **}**	Summary	Number of processed data records If the service is finished with an error, then the output is 0. In this case, the label **processing_result** gets the value **failed**. If no data records are processed, the label **processing_result** gets the value **empty**, otherwise **processed**.
wsp.services.subset.records.total_count/_max/_sum **{service_domain="ProcessValues",** **service_function="distributeStandard",** **distribution_id="***<from variable distributionId>***",** **processing_result="processed|empty|failed",** **device_id="***<client device ID>***",** **user_id="***<user ID>***",** **data_type="ProcessValues",** **record_type="remaining"** **}**	Summary	Number of process data waiting to be distributed. In mode **SINGLE_SERIAL**, this metric always includes the value 0. If the service is finished with an error, the label **processing_result** gets the value **failed.** If no data records were processed, the label **processing_result** gets the value **empty**, otherwise **processed**.

4.5.10 ProcessValues.importFromPccFiles

The following service-specific metrics are provided by the service **ProcessValues.importFromPccFiles:**

Metric name with labels	Type	Value

`wsp.services.subset.records.total_count/_max/_sum {service_domain="ProcessValues", service_function="importFromPccFiles", distribution_id="<from variable distributionId>", processing_result="processed|empty|failed", device_id="<client device ID>", user_id="<user ID>", data_type="ProcessValues", record_type="processed" }`	Summary	Number of processed data records If the service is finished with an error, then the output is 0. In this case, the label **processing_result** gets the value **failed**. If no data records are processed, the label **processing_result** gets the value **empty**, otherwise **processed**.
`wsp.services.subset.files.total_count/_max/_sum {service_domain="ProcessValues", service_function="importFromPccFiles", distribution_id"="<from variable distributionId>", device_id="<client device ID>", user_id="<user ID>", file_type="processed" }`	Summary	Number of processed files If the service is finished with an error, then the output is 0. In this case, the label **processing_result** gets the value **failed**. If no data records are processed, the label **processing_result** gets the value **empty**, otherwise **processed**.
`wsp.services.subset.files.total_count/_max/_sum {service_domain="ProcessValues", service_function="importFromPccFiles", distribution_id="<from variable distributionId>", processing_result="processed|empty|failed", device_id="<client device ID>", user_id="<user ID>",`	Summary	Number of files waiting to be processed. If the service is finished with an error, the label **processing_result** gets the value **failed.** If no data records were processed, the label **processing_result** gets the value **empty**, otherwise

file_type="remaining" **}**		**processed**.

4.5.11 ProcessValues.insert

The following service-specific metrics are provided by the service **ProcessValues.insert**:

Metric name with labels	Type	Value
wsp.services.subset.records.total_count/_max/_sum **{service_domain="ProcessValues",** **service_function="insert",** **distribution_id="***<content of the (optional) service parameter* ***distribution.id***>", **processing_result="processed|failed",** **device_id="***<client device ID>*", **user_id="***<user ID>*", **data_type="ProcessValues",** **record_type="inserted"** **}**	Summary	Number of processed data records If the service is finished with an error, then the output is 0, otherwise it is 1. In this case, the label **processing_result** gets the value **failed**, otherwise **processed**.